1. In problems below, find the (real) eigenvalues and associated eigenvectors of the given matrix A. Find a basis for each eigenspace of dimension 2 or larger.
   (a) $\begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$ **(b)** $\begin{bmatrix} 5 & -6 \\ 3 & -4 \end{bmatrix}$
   (c) $\begin{bmatrix} 10 & -8 \\ 6 & -4 \end{bmatrix}$ **(d)** $\begin{bmatrix} 7 & -6 \\ 12 & -10 \end{bmatrix}$
   (e) $\begin{bmatrix} 2 & 0 & -1 \\ 2 & -2 & -1 \\ -2 & 6 & 3 \end{bmatrix}$ **(f)** $\begin{bmatrix} 3 & 5 & -2 \\ 0 & 2 & 0 \\ 0 & 2 & 1 \end{bmatrix}$
   (g) $\begin{bmatrix} 1 & 0 & 0 \\ -6 & 8 & 2 \\ 12 & -15 & -3 \end{bmatrix}$

2. Find the eigenvalues and a basis for each eigenspace of the linear operator defined by the stated formula.
   [Hint: Work with the standard matrix for the operator.]
   (a) $T(x, y) = (x + 4y,\ 2x + 3y)$.
   (b) $T(x, y, z) = (2x - y - z,\ x - z,\ -x + y + 2z)$.

3. Suppose that λ is an eigenvalue of the matrix A with associated eigenvector $\mathbf{v}$ and that n is a positive integer. Show that λ^n is an eigenvalue of A^n with associated eigenvector $\mathbf{v}$.

4. Prove that the characteristic equation of a 2×2 matrix A can be expressed as $\lambda^2 - \operatorname{tr}(A)\,\lambda + \det(A) = 0$, where $\operatorname{tr}(A)$ is the trace of A.

5. Show that if
$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$
then the solutions of the characteristic equation of A are $\lambda = \frac{1}{2}\left[(a + d) \pm \sqrt{(a - d)^2 + 4bc}\right]$. Use this result to show that A has
   (a) two distinct real eigenvalues if $(a - d)^2 + 4bc > 0$.
   (b) two repeated real eigenvalues if $(a - d)^2 + 4bc = 0$.
   (c) complex conjugate eigenvalues if $(a - d)^2 + 4bc < 0$.

6. Let A be the matrix in Problem 5. Show that if $b \neq 0$, then $\mathbf{x}_1 = \begin{bmatrix} -b \\ a - \lambda_1 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} -b \\ a - \lambda_2 \end{bmatrix}$ are eigenvectors of A that correspond, **respectively**, to the eigenvalues
$$\lambda_1 = \tfrac{1}{2}\left[(a + d) + \sqrt{(a - d)^2 + 4bc}\right]$$
and
$$\lambda_2 = \tfrac{1}{2}\left[(a + d) - \sqrt{(a - d)^2 + 4bc}\right].$$

7. Prove: If λ is an eigenvalue of an invertible matrix A and $\mathbf{x}$ is a corresponding eigenvector, then $1/\lambda$ is an eigenvalue of A^{-1} and $\mathbf{x}$ is a corresponding eigenvector.

8. Prove: If λ is an eigenvalue of A, $\mathbf{x}$ is a corresponding eigenvector, and s is a scalar, then $\lambda - s$ is an eigenvalue of $A - sI$ and $\mathbf{x}$ is a corresponding eigenvector.

9. Prove: If λ is an eigenvalue of A and $\mathbf{x}$ is a corresponding eigenvector, then $s\lambda$ is an eigenvalue of sA for every scalar s and $\mathbf{x}$ is a corresponding eigenvector.

10. Find the eigenvalues and bases for the eigenspaces of
$$A = \begin{bmatrix} -2 & 2 & 3 \\ -2 & 3 & 2 \\ -4 & 2 & 5 \end{bmatrix},$$
and then use results from Problems 7 and 8 to find the eigenvalues and bases for the eigenspaces of
   (a) A^{-1}, **(b)** $A - 3I$, **(c)** $A + 2I$

11. Prove that the characteristic polynomial of an $n \times n$ matrix A has degree n and that the coefficient of λ^n in that polynomial is 1.

12. (a) Prove that if A is a square matrix, then A and A^T have the same eigenvalues. *[Hint: Look at the characteristic equation* $\det(\lambda I - A) = 0$.*]*
    (b) Show that A and A^T need not have the same eigenspaces. *[Hint: Use the result in Problem 6 to find a 2×2 matrix for which A and A^T have different eigenspaces.]*

13. In the following, show that A and B are not similar matrices.
    (a) $A = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix}$.
    (b) $A = \begin{bmatrix} 4 & -1 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ 2 & 4 \end{bmatrix}$.
    (c) $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
    (d) $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 3 & 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

14. Find a matrix P that diagonalizes A, and check your work by computing $P^{-1}AP$.
    (a) $A = \begin{bmatrix} 1 & 0 \\ 6 & -1 \end{bmatrix}$. **(b)** $A = \begin{bmatrix} -14 & 12 \\ -20 & 17 \end{bmatrix}$.
    (c) $A = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. **(d)** $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

15. Let $A = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix}$.
    (a) Find the eigenvalues of A.
    (b) For each eigenvalue λ, find the rank of the matrix $\lambda I - A$.
    (c) Is A diagonalizable? Justify your conclusion.

16. Find the geometric and algebraic multiplicity of each eigenvalue of the matrix A, and determine whether A is diagonalizable. If A is diagonalizable, then find a matrix P that diagonalizes A, and find $P^{-1}AP$.
    (a) $A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$. **(b)** $A = \begin{bmatrix} 19 & -9 & -6 \\ 25 & -11 & -9 \\ 17 & -9 & -4 \end{bmatrix}$.
    (c) $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 3 & 0 & 1 \end{bmatrix}$. **(d)** $A = \begin{bmatrix} 5 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & 5 \end{bmatrix}$.

17. In each part of the following, the characteristic equation of a matrix A is given. Find the size of the matrix and the possible dimensions of its eigenspaces.

(a) $(\lambda-1)(\lambda+3)(\lambda-5)=0$

(b) $\lambda^2(\lambda-1)(\lambda-2)^3=0$

(c) $\lambda^3(\lambda^2-5\lambda-6)=0$

(d) $\lambda^3-3\lambda^2+3\lambda-1=0$

18. Compute the matrix A^{10}.
(a) $A=\begin{bmatrix}0 & 3\\ 2 & -1\end{bmatrix}$. **(b)** $A=\begin{bmatrix}1 & 0\\ -1 & 2\end{bmatrix}$

19. Let $A=\begin{bmatrix}-1 & 7 & -1\\ 0 & 1 & 0\\ 0 & 15 & -2\end{bmatrix}$, $P=\begin{bmatrix}1 & 1 & 1\\ 0 & 0 & 1\\ 1 & 0 & 5\end{bmatrix}$.
Confirm that P diagonalizes A, and then compute A^{11}.

20. If A, B, and C are $n\times n$ matrices such that A is similar to B and B is similar to C, do you think that A must be similar to C? Justify your answer.

21. (a) Is it possible for an $n\times n$ matrix to be similar to itself? Justify your answer.
    (b) What can you say about an $n\times n$ matrix that is similar to $0_{n\times n}$? Justify your answer.
    (c) Is it possible for a nonsingular matrix to be similar to a singular matrix? Justify your answer.

22. Suppose that the characteristic polynomial of some matrix A is found to be

$$p(\lambda)=(\lambda-1)(\lambda-3)^2(\lambda-4)^3.$$

In each part, answer the question and explain your reasoning.

(a) What can you say about the dimensions of the eigenspaces of A?

(b) What can you say about the dimensions of the eigenspaces if you know that A is diagonalizable?

(c) If $\{\mathbf{v}_1,\mathbf{v}_2,\mathbf{v}_3\}$ is a linearly independent set of eigenvectors of A, all of which correspond to the same eigenvalue of A, what can you say about that eigenvalue?

23. In the following, find the standard matrix A for the given linear operator, and determine whether that matrix is diagonalizable. If diagonalizable, find a matrix P that diagonalizes A.

(a) $T(x_1,x_2)=(2x_1-x_2,\ x_1+x_2)$.

(b) $T(x_1,x_2)=(-x_2,\ -x_1)$.

(c) $T(x_1,x_2,x_3)=(8x_1+3x_2-4x_3,-3x_1+x_2+3x_3,\ 4x_1+3x_2)$.

(d) $T(x_1,x_2,x_3)=(3x_1,\ x_2,\ -x_2)$.

24. If P is a fixed $n\times n$ matrix, then the similarity transformation $A\mapsto P^{-1}AP$ can be viewed as an operator $S_P(A)=P^{-1}AP$ on the vector space M_n of $n\times n$ matrices.

(a) Show that S_P is a linear operator.

(b) Find the kernel of S_P.

(c) Find the rank of S_P.

25. For each of the following, find the distinct singular values of A.
(a) $A=\begin{bmatrix}1 & 2 & 0\end{bmatrix}$ **(b)** $A=\begin{bmatrix}3 & 0\\ 0 & 4\end{bmatrix}$
(c) $A=\begin{bmatrix}1 & -2\\ 2 & 1\end{bmatrix}$ **(d)** $A=\begin{bmatrix}\sqrt{2} & 0\\ 1 & \sqrt{2}\end{bmatrix}$

26. For each of the following, find a singular value decomposition of A.
(a) $A=\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}$ **(b)** $A=\begin{bmatrix}-3 & 0\\ 0 & -4\end{bmatrix}$
(c) $A=\begin{bmatrix}4 & 6\\ 0 & 4\end{bmatrix}$ **(d)** $A=\begin{bmatrix}3 & 3\\ 3 & 3\end{bmatrix}$
(e) $A=\begin{bmatrix}-2 & 2\\ -1 & 1\\ 2 & -2\end{bmatrix}$ **(f)** $A=\begin{bmatrix}-2 & -1 & 2\\ 2 & 1 & -2\end{bmatrix}$
(g) $A=\begin{bmatrix}1 & 0\\ 1 & 1\\ -1 & 1\end{bmatrix}$ **(h)** $A=\begin{bmatrix}6 & 4\\ 0 & 0\\ 4 & 0\end{bmatrix}$